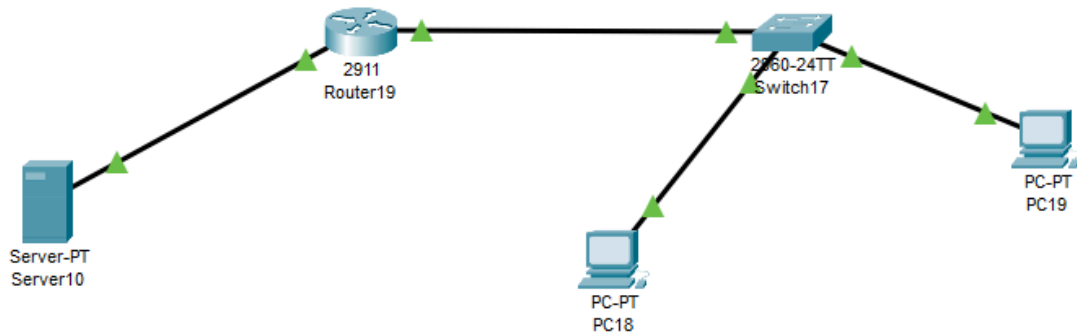


DHCP Connection

This is the main project structure.

PC1, PC2 -> Switch1 -> Router19 <- Server10 (DHCP server)



I implemented a small DHCP server setup, which was moderately challenging but very insightful.

First, I connected the DHCP server to the router and configured the necessary DHCP settings. This included defining the address pool for the client PCs located on a different network segment.

DHCP Configuration:

I created a dedicated IP address pool for the client machines on the second network to ensure proper dynamic IP allocation.

DHCP settings:

IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

Also setup a pool for the PCs on the other Network.

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.20.1

DNS Server: 0.0.0.0

Start IP Address: 192 168 20 10

Subnet Mask: 255 255 255 0

Maximum Number of Users: 246

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	0.0.0.0	192.168....	255.255....	246	0.0.0.0	0.0.0.0

Router Configuration (ROUTER19):

The router was configured as follows:

```
en
conf t
int g0/0

ip address 192.168.10.1 255.255.255.0
no shutdown

int g0/1
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.10.2
no shutdown
```

The ip helper-address command was used on the second interface to forward DHCP requests to the DHCP server located in the 192.168.10.0/24 network.

Final Setup:

After completing the router configuration, I connected the switch along with PCs (PC18 and PC19). Once the network setup was complete, I enabled DHCP on the client machines. The devices successfully received IP addresses from the DHCP server, confirming that the configuration was working correctly.

-> Sk. Md. Rashid Assef Shibly